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/* Name-Manaswi Butle
/PRN-25070521224/

SECTION-D-II

Value Added Problem-2 */

#include <stdio.h>
#include <string.h>

#define MAX 1000000

char stack[MAX];
int top = -1;

// Push function
void push(char c) {
    stack[++top] = c;
}

// Pop function
char pop() {
    if (top == -1)
        return '\0';
    return stack[top--];
}

// Check if matching pair
int isMatchingPair(char open, char close) {
    if (open == '(' && close == ')') return 1;
    if (open == '{' && close == '}') return 1;
    if (open == '[' && close == ']') return 1;
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        return 0;
    }

int isBalanced(char *s) {
    for (int i = 0; s[i] != '\0'; i++) {

        // If opening bracket, push
        if (s[i] == '(' || s[i] == '{' || s[i] == '[') {
            push(s[i]);
        }

        // If closing bracket
        else if (s[i] == ')' || s[i] == '}' || s[i] == ']') {

            if (top == -1) // No matching opening
                return 0;

            char topElement = pop();

            if (!isMatchingPair(topElement, s[i]))
                return 0;
        }
    }

    // If stack is empty → balanced
    return (top == -1);
}

int main() {
    printf("Name :Manaswi Butle \n");
}

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printf("PRN :25070521224\n");

printf("Section :D-II\n");

printf("Value Added Problem : 2\n");

printf("-----\n");

char s[MAX];

printf("Enter expression: ");

scanf("%s", s);

if (isBalanced(s))

    printf("true\n");

else

    printf("false\n");

return 0;
}

```

main.c	Run	Output
<pre> 1 /* Name-Manaswi Butle 2 /PRN-25070521224/ 3 SECTION-D-II 4 Value Added Problem-2 */ 5 #include <stdio.h> 6 #include <string.h> 7 8 #define MAX 1000000 9 10 char stack[MAX]; 11 int top = -1; </pre>		<pre> Name :Manaswi Butle PRN :25070521224 Section :D-II Value Added Problem : 2 ----- Enter expression: {[[]]} true === Code Execution Successful === </pre>